KAIZEN MATHS

Lesson starter

Binomial and Geometric Distributions

Geometric

Name _____

S1

X ~ Geo(0.4). Find P(X = 5), where X is the trial of the first success.

S2

X ~ Geo(0.15). Find P(X ≤ 5).

S3

X ~ Geo(0.6). Find P(X ≥ 10).

S4

X ~ Geo(0.5). Find E(X) and Var(X).

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Exit ticket

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X ~ Geo(0.7). Find P(X = 7), where X is the trial of the first success.

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X ~ Geo(0.45). Find P(X ≤ 7).

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Teacher answers & working

Binomial and Geometric Distributions · Geometric

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S1

X ~ Geo(0.4). Find P(X = 5), where X is the trial of the first success.

Answer:

0.0518

Working:

For the first success on trial 5, there are 4 failures first.

Use $P(X = r) = (1 - p)^{r-1}p$.

$P(X = 5) = (0.6)^4(0.4)$.

$P(X = 5) \approx 0.0518$.

0.0518

S2

X ~ Geo(0.15). Find P(X ≤ 5).

Answer:

0.5563

Working:

P(X ≤ r) means the first success happens by trial 5.

Use $P(X \leq r) = 1 - (1 - p)^r$.

$P(X \leq 5) = 1 - (0.85)^5$.

$P(X \leq 5) \approx 0.5563$.

0.5563

S3

X ~ Geo(0.6). Find P(X ≥ 10).

Answer:

0.0003

Working:

X ≥ 10 means no success in the first 9 trials.

$P(X \geq 10) = (1 - p)^9$.

$P(X \geq 10) = (0.4)^9 \approx 0.0003$.

0.0003

Teacher answers & working

Binomial and Geometric Distributions · Geometric

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S4

X ~ Geo(0.5). Find E(X) and Var(X).

Answer:

$E(X) = 2$, $Var(X) = 2$

Working:

For $X \sim Geo(p)$, $E(X) = 1 / p$ and $Var(X) = (1 - p) / p^2$.

$E(X) = 1 / 0.5 = 2$.

$Var(X) = 0.5 / (0.5)^2 = 2$.

$E(X) = 2$, $Var(X) = 2$

E1

X ~ Geo(0.7). Find P(X = 7), where X is the trial of the first success.

Answer:

0.0005

Working:

For the first success on trial 7, there are 6 failures first.

Use $P(X = r) = (1 - p)^{r - 1}p$.

$P(X = 7) = (0.3)^6(0.7)$.

$P(X = 7) \approx 0.0005$.

0.0005

E2

X ~ Geo(0.45). Find P(X ≤ 7).

Answer:

0.9848

Working:

P(X ≤ r) means the first success happens by trial 7.

Use $P(X \leq r) = 1 - (1 - p)^r$.

$P(X \leq 7) = 1 - (0.55)^7$.

$P(X \leq 7) \approx 0.9848$.

0.9848

Teacher answers & working

Binomial and Geometric Distributions · Geometric

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E3

X ~ Geo(0.35). Find P(X ≥ 7).

Answer:

0.0754

Working:

X ≥ 7 means no success in the first 6 trials.

$P(X \geq 7) = (1 - p)^6$.

$P(X \geq 7) = (0.65)^6 \approx 0.0754$.

0.0754